

[Part 1] IMTD Signal Generation & Cryptographic Initialization (CNAV-2 Format)

[Init] Generating ECDSA (secp256r1) Key Pair...

[Tx] Pure Pilot Channel generated (No Puncturing).

[Tx] Generating TESLA Hash Chain (n=10)...

[Tx] Constructing nav5 with IMTD Payload (MAC at SoM, Key at EoM)...

[Tx] IMTD 1 Epoch (5 Frames, 9000 Symbols) successfully assembled in CNAV-2 format.

[Tx] Generating Baseband Chips (Chunking Preparation)...

[Part 2] Coarse Signal Acquisition (1st Symbol Only)

[Part 3] Signal Tracking (Symbol-by-Symbol Chunking)

Tracking 9210 symbols...

[1000 / 9210] processed...

[2000 / 9210] processed...

[3000 / 9210] processed...

[4000 / 9210] processed...

[5000 / 9210] processed...

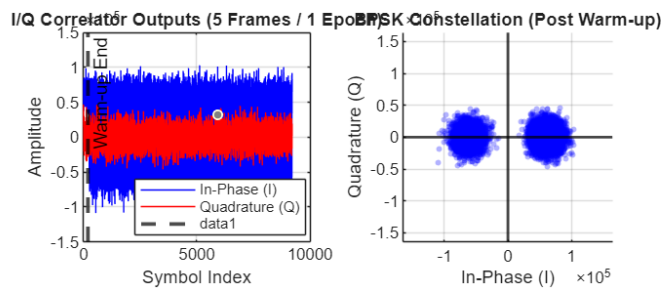
[6000 / 9210] processed...

[7000 / 9210] processed...

[8000 / 9210] processed...

[9000 / 9210] processed...

Tracking Done!



[Part 4] CNAV-2 Epoch Demodulation (nav1 ~ nav5)

[Rx] nav1: LDPC & CRC Passed.

[Rx] nav2: LDPC & CRC Passed.

[Rx] nav3: LDPC & CRC Passed.

[Rx] nav4: LDPC & CRC Passed.

[Rx] nav5 Sub-frame 3: CRC Passed!

[Rx] nav5 successfully parsed (MAC and Key extracted).

[Part 5] Cryptographic Verification (IMTD)

[Rx] 1. TESLA Key Chain Verify : PASS (Hash matched stored commitment K_3)

[Rx] 2. Navigation MAC Verify : PASS (nav1~nav4 are Authentic & Intact)

[Rx] 3. Root Signature Verify : PASS (ECDSA-512 Validates Derived K_0)

[+++] IMTD Cryptographic Authentication Fully Successful! Epoch is Genuine. [+++]